HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed precise technical drawings for **solar PV systems**, including rooftop and ground-mounted installations, with a focus on domestic properties, ensuring compliance with Irish and EU standards.
- Interpreted data from site surveys, translating it into accurate **design plans** and preparing detailed **wiring diagrams** and technical specifications for project deployment.
- Collaborated with sales teams and project managers to ensure project delivery within specified timelines and budget constraints, facilitating smooth transitions from design to installation.
- Designed and implemented **Java Spring Boot REST APIs** for cross-service data exchange, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Managed production support and incident response for enterprise applications, analyzing logs via **Splunk** and **Dynatrace** to resolve P1/P2 issues within strict SLAs.
- Automated data extraction and reporting tasks using **Python scripting**, processing approximately **15,000 records** per batch for upstream analytical consumption.
- Containerised Java applications using **Docker** and deployed to **AWS EC2** instances, configuring basic resource limits for predictable performance during rolling updates.

SKILLS

Solar PV Design – Solar PV systems, Rooftop installations, Ground-mounted installations, Wiring diagrams, Technical specifications, Site surveys

Engineering Software – AutoCAD, Helioscope, PVSyst, PVSol, HOMER

Programming Languages – Java, Python, SQL

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Docker, Jenkins, Git, Jira, Confluence, Agile, Scrum

PROJECTS

Solar PV System Design for Domestic Properties

- Created detailed technical drawings and wiring schematics for residential solar PV installations, integrating site-specific survey data.
- Prepared comprehensive Bill of Materials (BOM) and technical specifications, ensuring adherence to national and EU renewable energy regulations.
- Collaborated with project managers to refine designs based on installation team feedback, optimizing for efficient deployment and system performance.

Automated ETL Pipeline for Financial Transaction Data

- Developed Python scripts to extract, transform, and load financial transaction data from PostgreSQL into an analytical data store, processing ~**2,800 transactions** per hour.
- Implemented data validation checks and error handling within the ETL process, reducing data quality issues by an estimated **30%** for downstream reporting.
- Scheduled and monitored ETL jobs using cron on Linux, ensuring timely delivery of processed data for operational dashboards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)